

Exercise 3 - Theoretical question #2: the resolution of the identity (density fitting)

The bottleneck

In HF and DFT the Coulomb and exchange matrices are built from four-centre two-electron integrals:

$$J_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\nu|\lambda\sigma), \quad K_{\mu\nu} = \sum_{\lambda\sigma} P_{\lambda\sigma} (\mu\lambda|\nu\sigma), \quad (\mu\nu|\lambda\sigma) = \iint \phi_\mu(1) \phi_\nu(1) r_{12}^{-1} \phi_\lambda(2) \phi_\sigma(2) dr_1 dr_2.$$

The number of these integrals grows as N^4 , where N is the number of basis functions. For 2-chlorophenol with def2-TZVP ($N = 284$, Exercise 1) there are about 8.1×10^8 unique integrals, roughly 6.5 GB, and they must be recalculated in every SCF iteration or read back from disk. Correlated methods such as MP2 and coupled cluster also have to transform them to the molecular orbital basis, which scales as N^5 .

The idea

A four-centre integral is the Coulomb repulsion between two charge distributions, the orbital products $\rho_{\mu\nu}(r) = \phi_\mu(r) \phi_\nu(r)$ and $\rho_{\lambda\sigma}(r)$. In the resolution of the identity (RI), also called density fitting, each product is expanded in an auxiliary basis of atom-centred functions χ_P :

$$\rho_{\mu\nu}(r) \approx \sum_P c_{\mu\nu}^P \chi_P(r).$$

The coefficients are found by minimising the Coulomb self-repulsion of the fitting error. This gives $c_{\mu\nu} = V^{-1} (P|\mu\nu)$ with the Coulomb metric $V_{PQ} = (P|Q)$, and the error in the energy is then only second order in the error of the fitted density. The four-index integral splits into three- and two-index integrals:

$$(\mu\nu|\lambda\sigma) \approx \sum_{PQ} (\mu\nu|P) [V^{-1}]_{PQ} (Q|\lambda\sigma).$$

This is the same as inserting an approximate identity operator, $1 \approx \sum_{PQ} |P\rangle [V^{-1}]_{PQ} \langle Q|$, between the two charge distributions, which is where the name comes from. The accuracy depends on the auxiliary basis. ORCA has matched auxiliary sets for different purposes, for example def2/J for Coulomb, def2/JK for Coulomb and exchange, def2-TZVP/C for correlation and SARC/J for relativistic basis sets, and AutoAux can generate one automatically.

Where the speed-up comes from

First, far fewer integrals are needed. Only three-centre $(\mu\nu|P)$ and two-centre $(P|Q)$ integrals appear, and the auxiliary basis is only 1.5–3 times larger than the orbital basis (def2/J: $N_{\text{aux}} = 449$ for 2-chlorophenol). The number of integrals drops from about $N^4/8 = 8.1 \times 10^8$ to $N^2 N_{\text{aux}}/2 = 1.8 \times 10^7$, around 45 times fewer. At about 145 MB they fit in memory, so they do not have to be recalculated or read from disk.

Second, the Coulomb matrix is obtained in two cheap steps (RI-J). The density is fitted first, $d = V^{-1} g$ with $g_Q = \sum_{\lambda\sigma} (Q|\lambda\sigma) P_{\lambda\sigma}$, and then $J_{\mu\nu} = \sum_P (\mu\nu|P) d_P$. Both steps cost $N^2 N_{\text{aux}}$ instead of N^4 and are dense matrix operations that run fast on modern processors. For a GGA functional like PBE, J is the

only two-electron term, so RI-J removes the main bottleneck. The fitting error is small and systematic and largely cancels in energy differences.

The exchange matrix couples the indices crosswise and does not factorise as well. ORCA either fits it too (RIJK, with a /JK auxiliary basis) or uses RIJCOSX, which is RI for J and the chain-of-spheres method for K. In COSX one electron coordinate is integrated analytically and the other on a numerical grid, $K_{\mu\nu} \approx \sum_g w_g \phi_\mu(r_g) \sum_{\lambda\sigma} A_{\nu\lambda}(r_g) P_{\lambda\sigma} \phi_\sigma(r_g)$ with $A_{\nu\lambda}(r_g) = \int \phi_\nu(r) \phi_\lambda(r) |r - r_g|^{-1} dr$, and the cost grows almost linearly for large molecules. In Exercise 1 the PBE runs used RI-J (def2/J) and the B3LYP runs RIJCOSX (def2/J).

The same factorisation speeds up correlated methods. In RI-MP2 the MO integrals are written as $(ia|jb) \approx \sum_p B_{ia}^p B_{jb}^p$, which avoids the expensive four-index transformation and makes the calculation an order of magnitude or more faster. ORCA's DLPNO methods are also built on it.